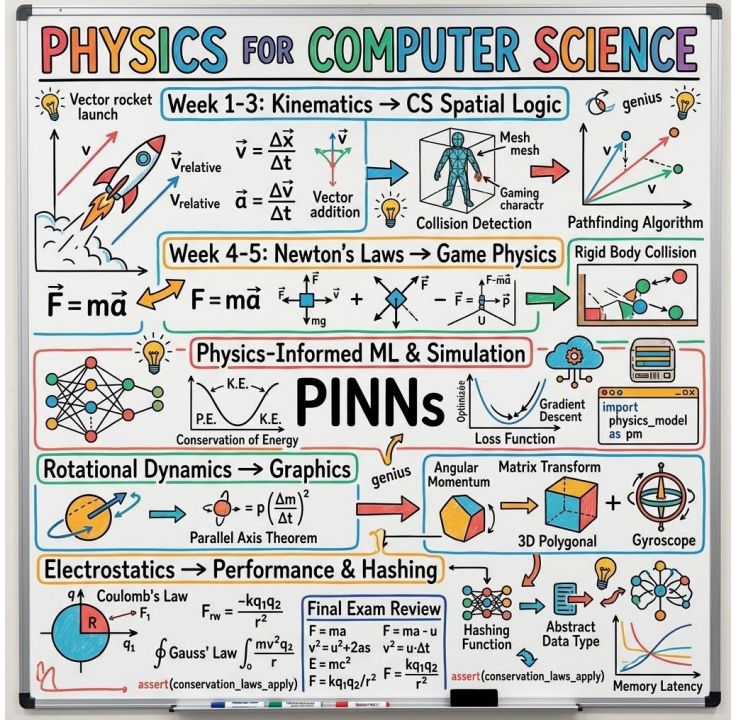
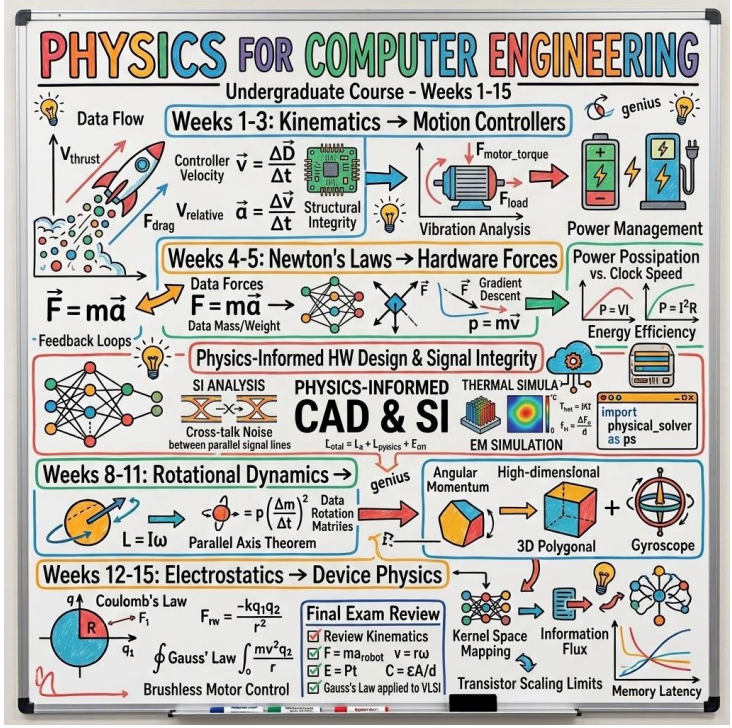
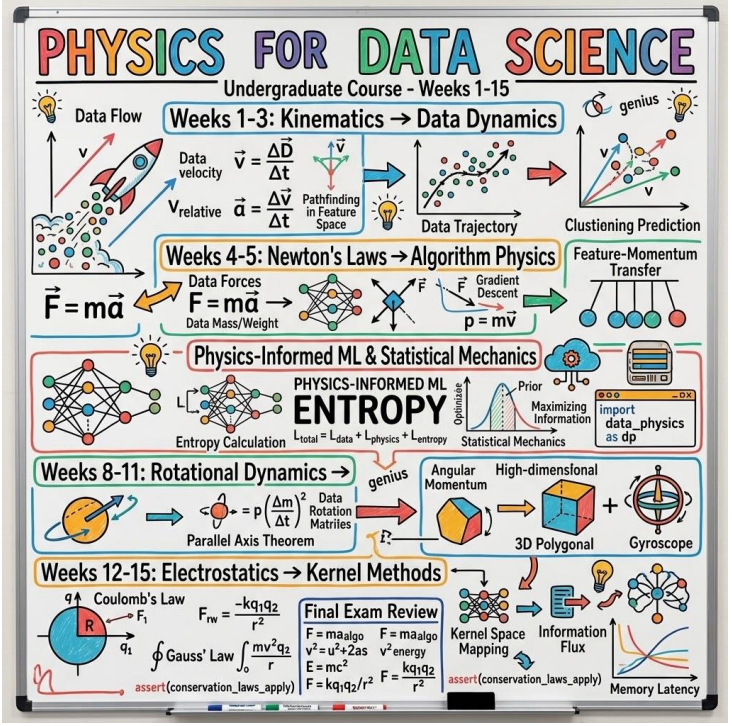
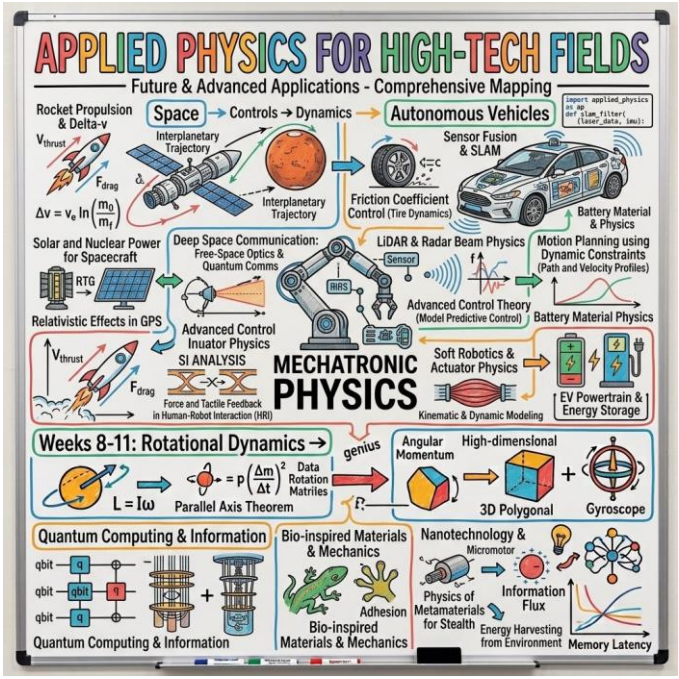








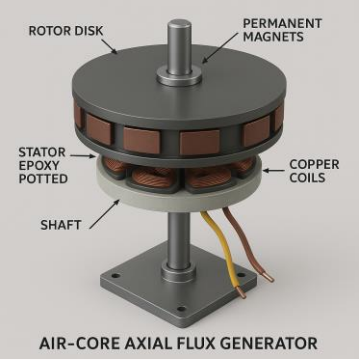
Applied Physics
PH101

Faculty of Basic Sciences

Week_08

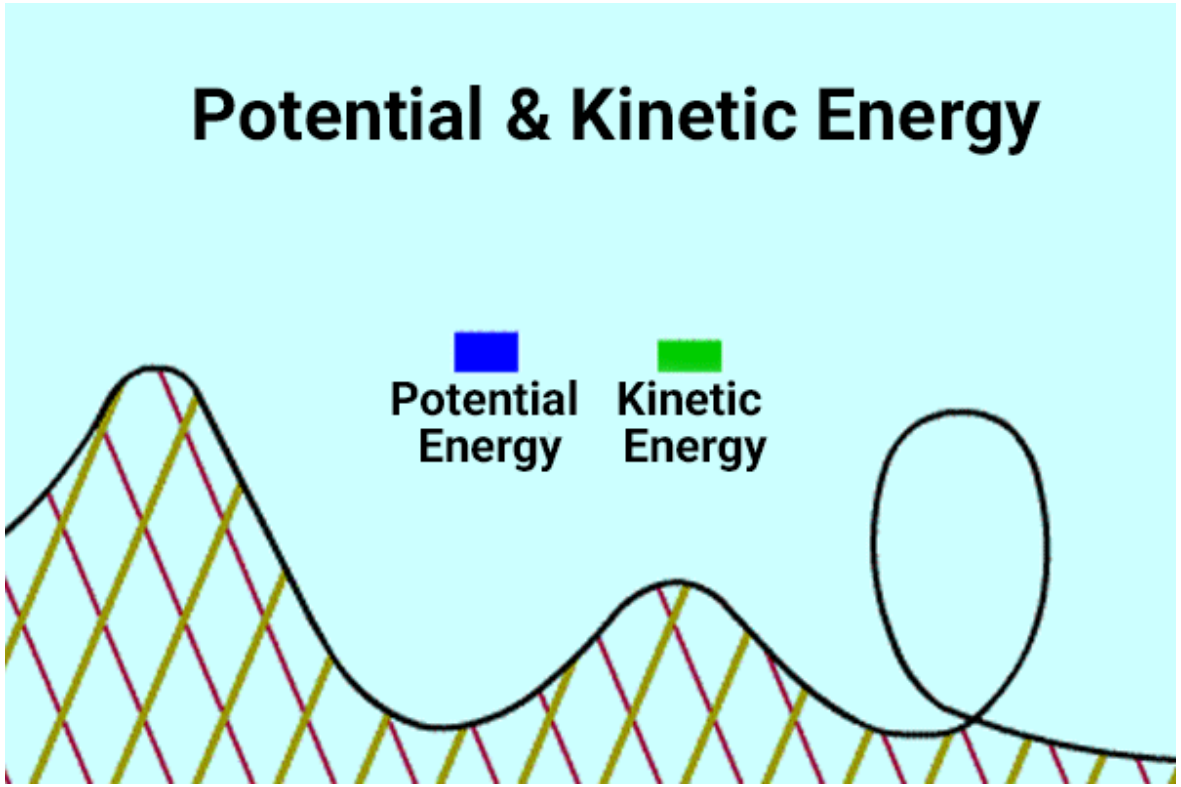
Fall_2026

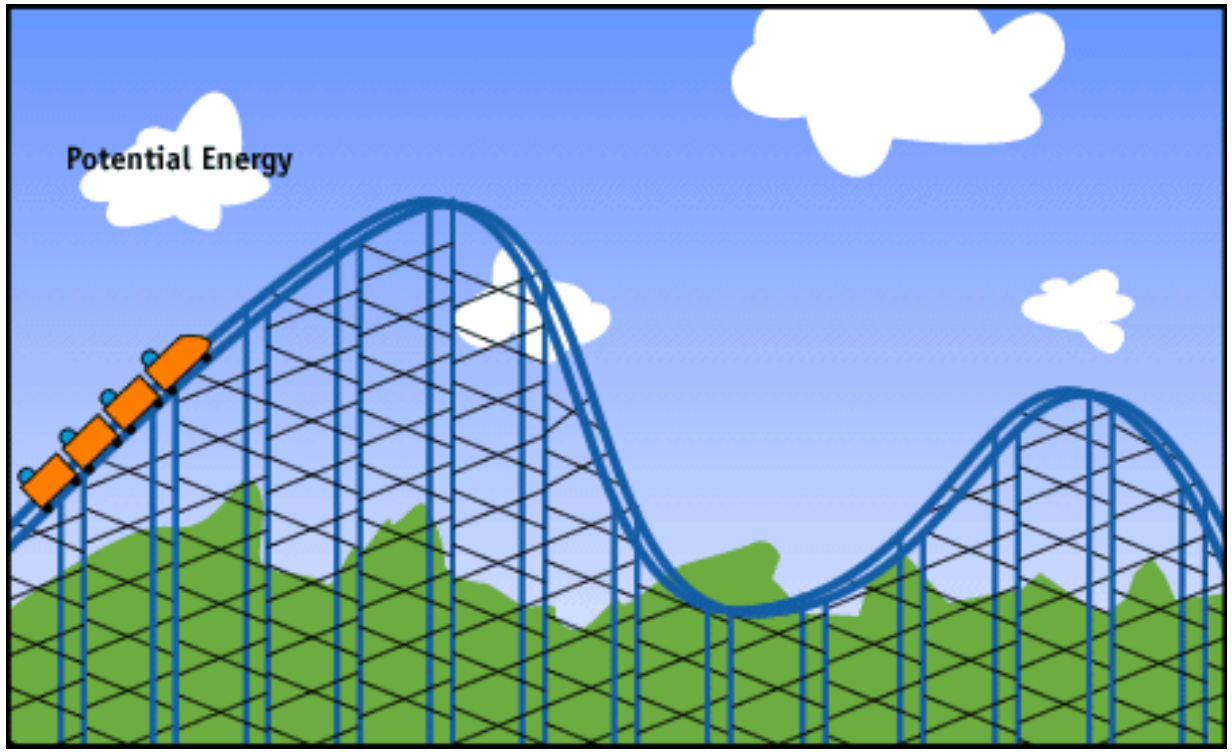
Weekly Plan



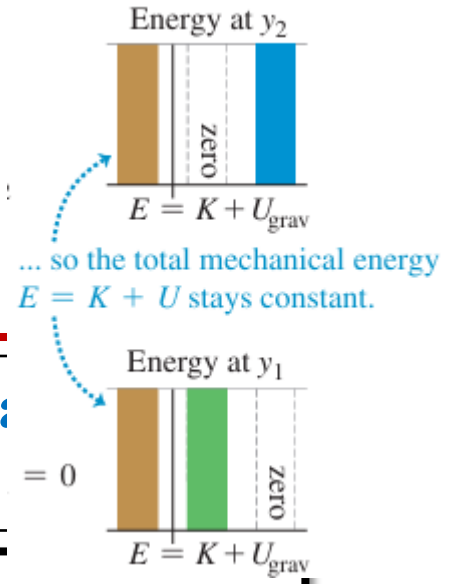
- **Work and Kinetic Energy**
  - Gravitational Potential Energy
  - Elastic Potential Energy
  - Conservative and Nonconservative Forces
  - Force and Potential Energy & Energy Diagrams

Conversion of Mechanical Energy





Example

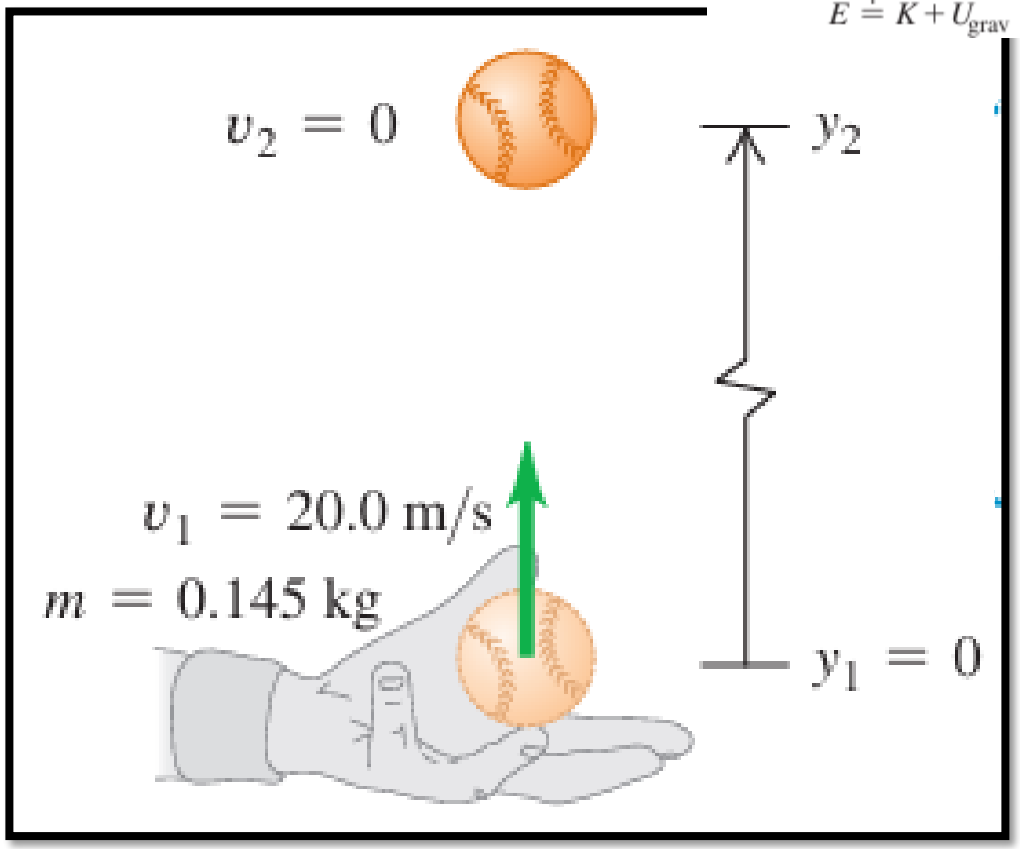


You throw a **0.145-kg** baseball straight up, giving it an **initial velocity of ma**
20.0 m/s. **Find how high** it goes, ignoring air resistance.

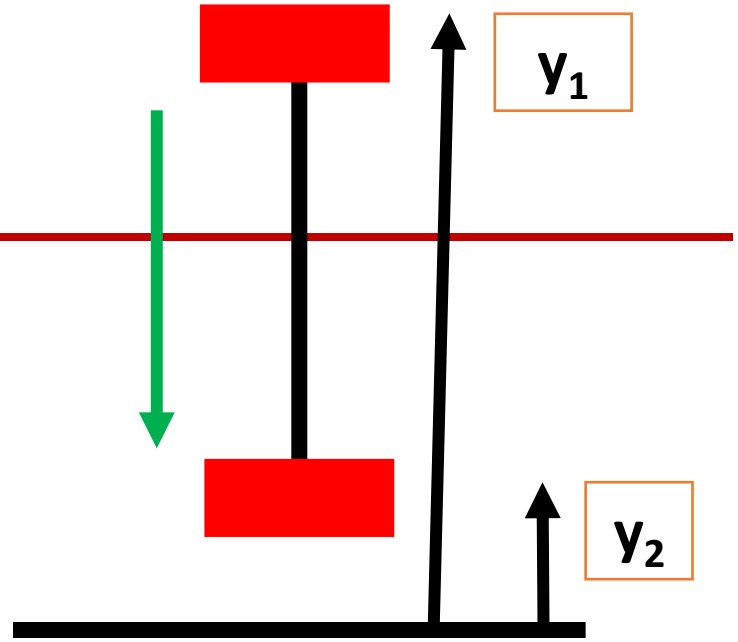
$$K_2 + U_2 = K_1 + U_1$$

Initial	Final
$KE_1 = \frac{1}{2}mV_1^2$	$KE_2 = 0$
$U_1 = 0$	$U_2 = mgy_2$

$$y_2 = 20.4\ \text{m}$$



Elastic Potential Energy



$$K_1 + U_1 = K_2 + U_2$$

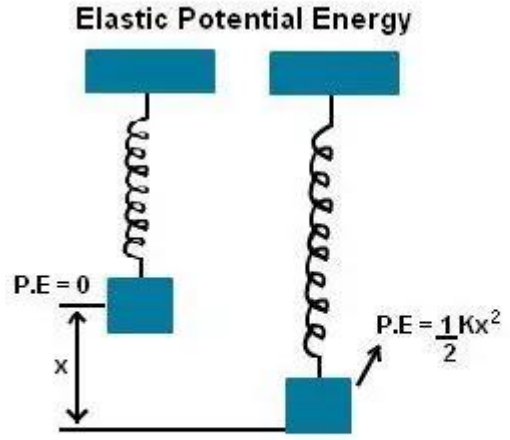
$$K_1 + U_{g1} = K_2 + U_{g2}$$

$$K_1 + U_{e1} = K_2 + U_{e2}$$

$$K_1 + U_{g1} + U_{e1} = K_2 + U_{g2} + U_{e2}$$

$$W_{other} + K_1 + U_{g1} + U_{e1} = K_2 + U_{g2} + U_{e2}$$

$$K_1 + U_{g1} + U_{e1} = W_{other} + K_2 + U_{g2} + U_{e2}$$



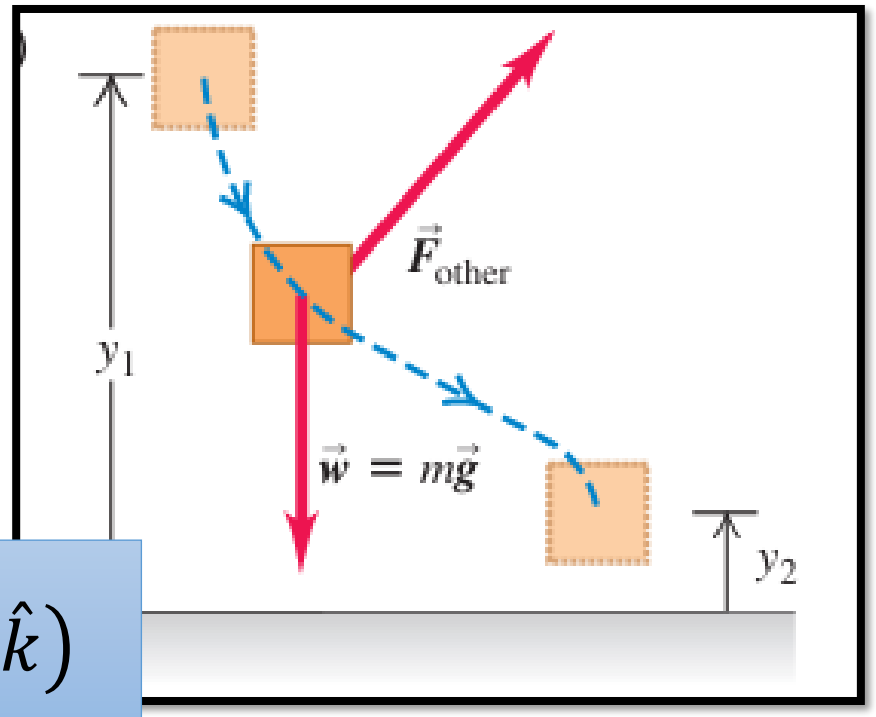
Gravitational Potential Energy for Motion Along a Curved Path

$$\bar{w} = m\bar{g}$$

$$\bar{w} = -mg\hat{j}$$

$$\Delta\bar{s} = \Delta x\hat{\imath} + \Delta y\hat{j}$$

$$Work = \bar{w}.\Delta\bar{s}$$



$$W = \int_1^2 \vec{F}_g \cdot d\vec{r} = \int_1^2 (-mg\hat{j}) \cdot (dx\,\hat{\imath} + dy\,\hat{j} + dz\,\hat{k})$$

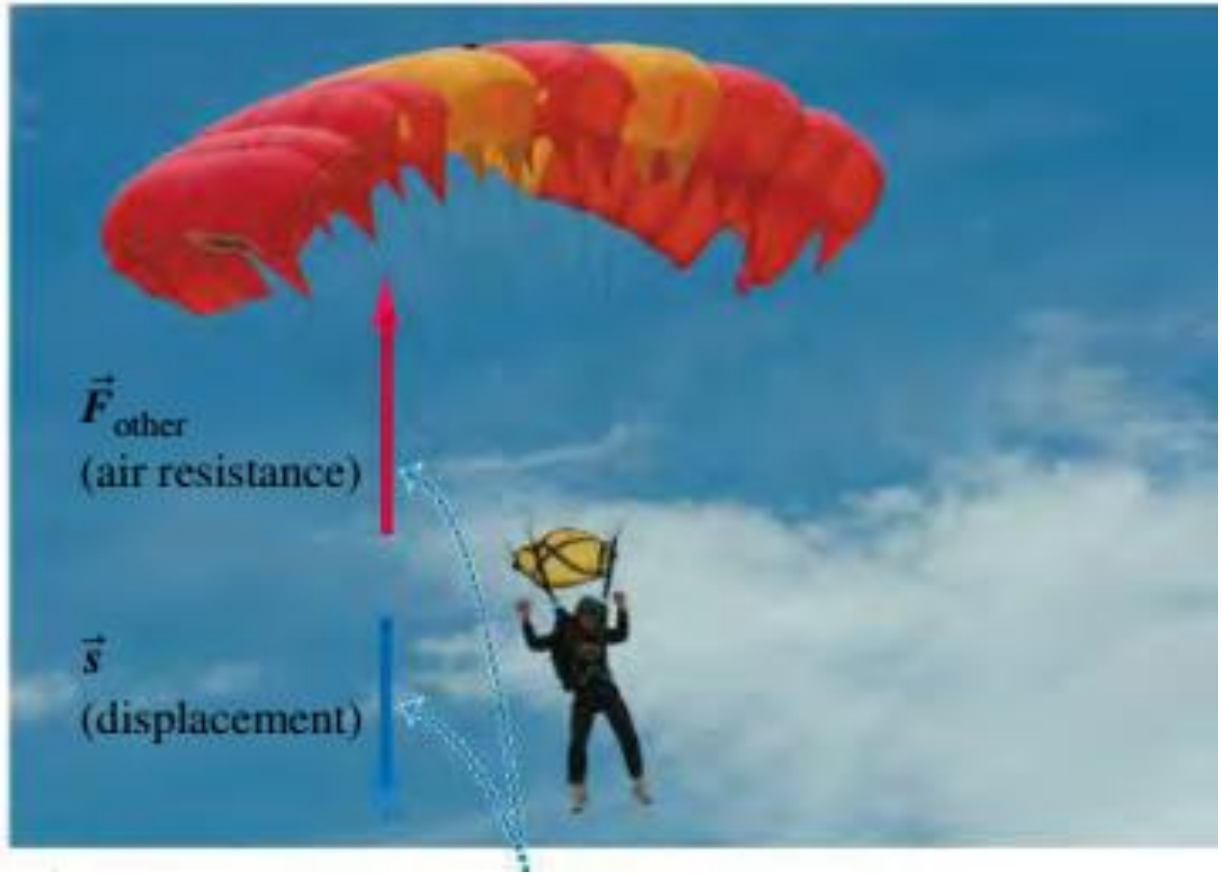
$$W = -mg\int_{y_1}^{y_2} d\,y = -mg(y_2 - y_1) = -(mgy_2 - mgy_1)$$

$$Work = -mg\,\Delta y$$

$$K_1 + U_{g1} = W_{other} + K_2 + U_{g2}$$

Conversion of Mechanical Energy

Parachutist moves downward at a constant speed, the upward force of air resistance does negative work W_{other} on him.
The total mechanical energy $E = K + U$ decreases.



- $\vec{F}_{\text{other}}$ and $\vec{s}$ are opposite, so $W_{\text{other}} < 0$.
- Hence $E = K + U_{\text{grav}}$ must decrease.
- The parachutist's speed remains constant, so K is constant.
- The parachutist descends, so U_{grav} decreases.

Conversion of Mechanical Energy

Conservation of total energy

Includes mechanical, thermal, chemical, electrical, sound, and other forms of energy within the chosen system.

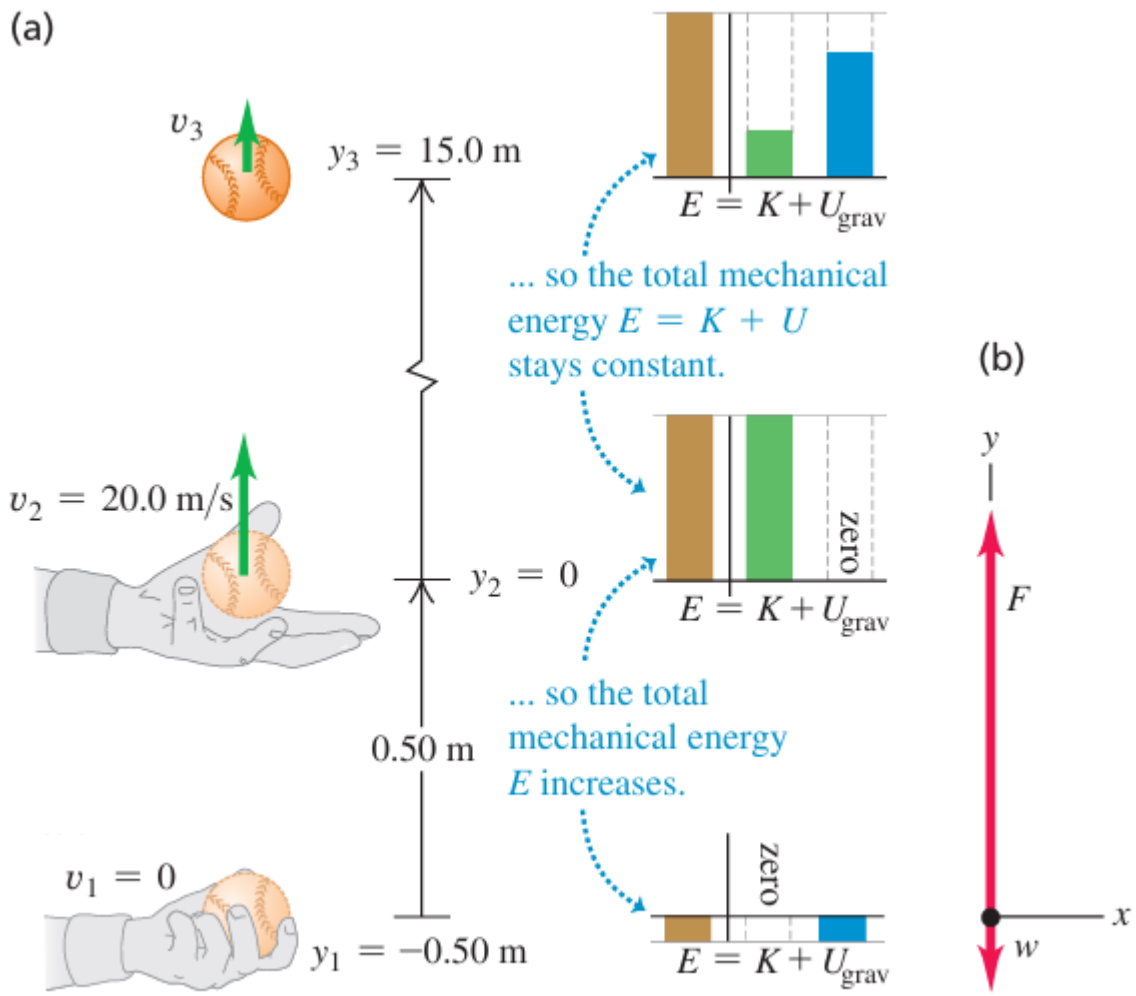
Mechanical energy is not always conserved

Friction, air resistance, inelastic deformation, and other dissipative processes can convert mechanical energy into internal energy or transfer it to the surroundings.

You throw a **0.145-kg** baseball straight up, suppose your hand moves **upward by 0.50 m** while you are throwing the ball. The ball leaves your hand with an **upward velocity of 20.0 m/s**. **(a) Find the magnitude of the force** (assumed constant) that your hand exerts on the ball. **(b) Find the speed of the ball at a point 15.0 m** above the point where it leaves your hand. Ignore air resistance.

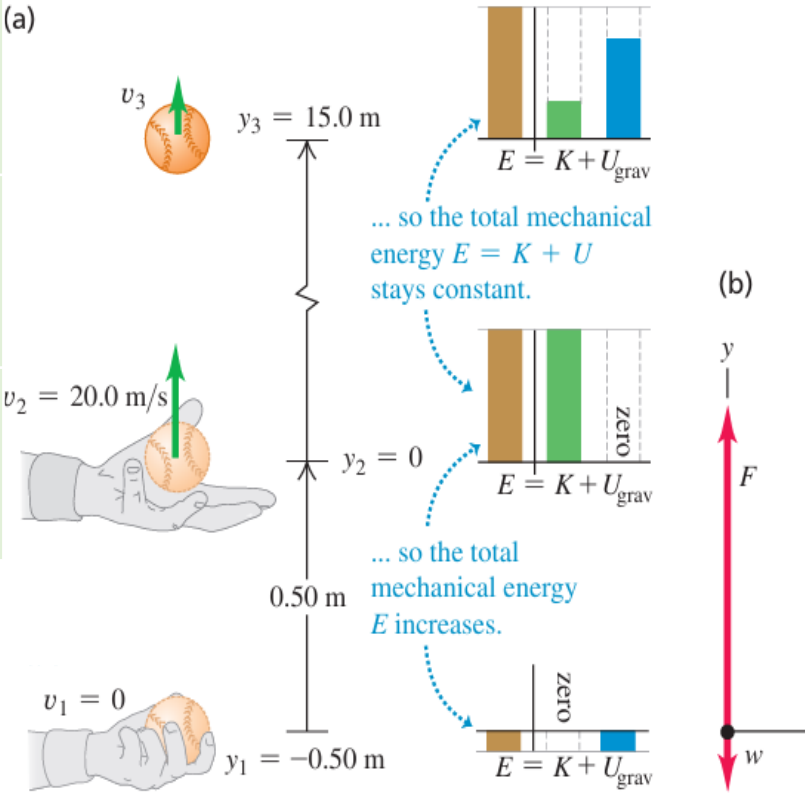
$$K_2 + U_2 = K_3 + U_3$$

$$W_{other} + K_1 + U_1 = K_2 + U_2$$



You throw a **0.145-kg** baseball straight up, suppose your hand moves **upward by 0.50 m** while you are throwing the ball. The ball leaves your hand with an **upward velocity of 20.0 m/s**. **(a) Find the magnitude of the force** (assumed constant) that your hand exerts on the ball. **(b) Find the speed of the ball at a point 15.0 m** above the point where it leaves your hand. Ignore air resistance.

Layer	What students must understand	Why it's hard
1. Two different phases	During the throw (hand contact) vs. after release (free fall)	two phases
2. Two forces during the throw	Hand pushes up, gravity pulls down — the net force is $F - mg$	gravity acts during the throw
3. The "gravity cost"	The hand must do work against gravity while accelerating the ball	hand only provides acceleration



(a) using only $F = ma$ and get $F = 58$N — then wonder why the answer is 59.4 N.
(b) The **1.4 N difference** (which is mg) is exactly what confuses them

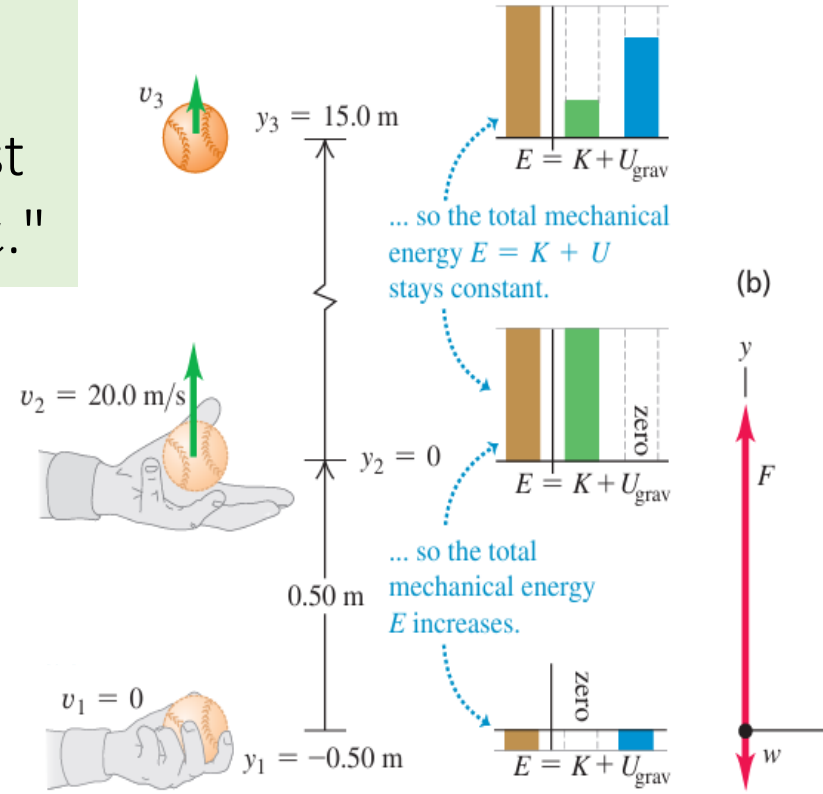
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Normal mental model: "The hand accelerates the ball, so $F = ma$. Gravity only matters after the ball leaves the hand."

Correct model: "Gravity acts *during* the throw too. The hand must overcome gravity *and* accelerate the ball. So $F_{\text{hand}} - mg = ma$."

"losses" concept — the hand's force must cover two jobs:

- **Support the ball's weight (the "gravity cost")**
- **Accelerate the ball to 20 m/s**



You throw a **0.145-kg** baseball straight up, suppose your hand moves **upward by 0.50 m** while you are throwing the ball. The ball leaves your hand with an **upward velocity of 20.0 m/s**. **(a) Find the magnitude of the force** (assumed constant) that your hand exerts on the ball. **(b) Find the speed of the ball at a point 15.0 m** above the point where it leaves your hand. Ignore air resistance.

"If you held the ball still in your hand and lifted it 0.50 m at constant speed, how much force would your hand exert?"

$F = mg = 0.145 \times 9.8 = 1.42\text{N}.$

"Now, if you throw it so it reaches 20 m/s over 0.50 m, is the force more or less than 1.42 N?"

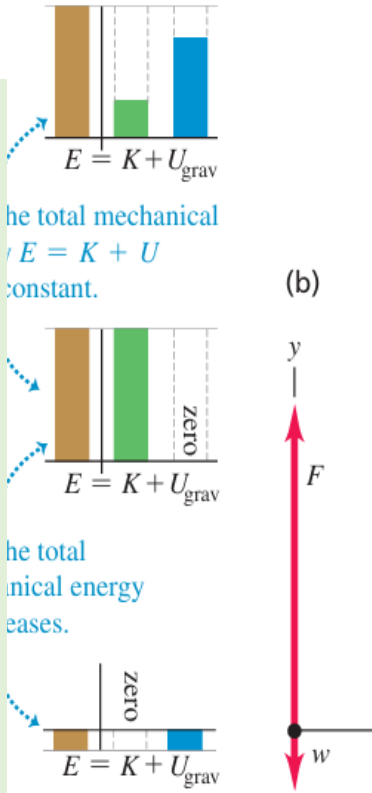
Answer: More — you need extra force to accelerate it.

"So the total force is the force to hold it plus the force to accelerate it."

$$\boldsymbol{F}_{\text{total}} = \underbrace{\boldsymbol{mg}}_{\text{hold it up}} + \underbrace{\boldsymbol{ma}}_{\text{accelerate it}}$$

This immediately gives:

$$F = 1.42 + 0.145 \times 400 = 1.42 + 58.0 = 59.4\ \text{N}$$



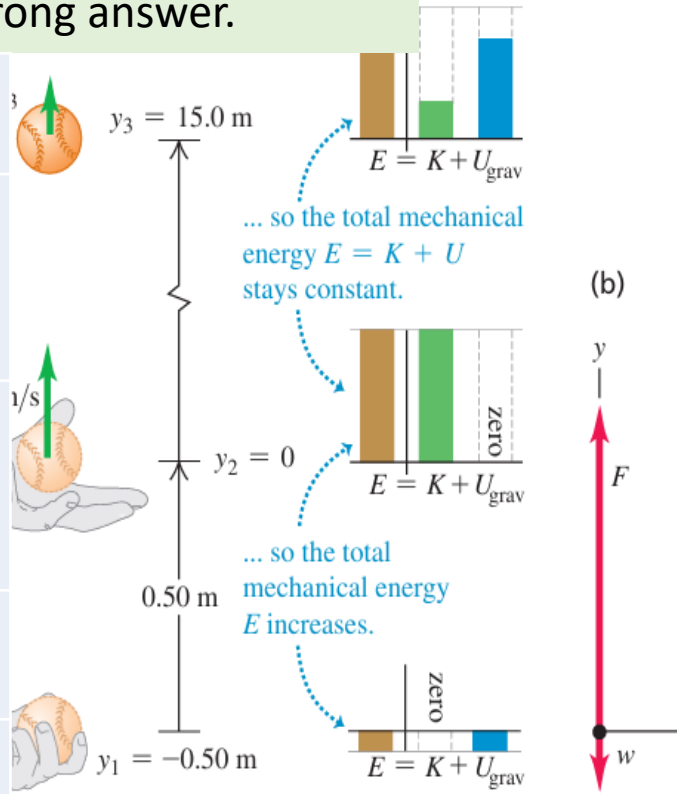
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Net force (upward positive):

$$F_{\text{net}} = F_{\text{hand}} - mg = ma$$

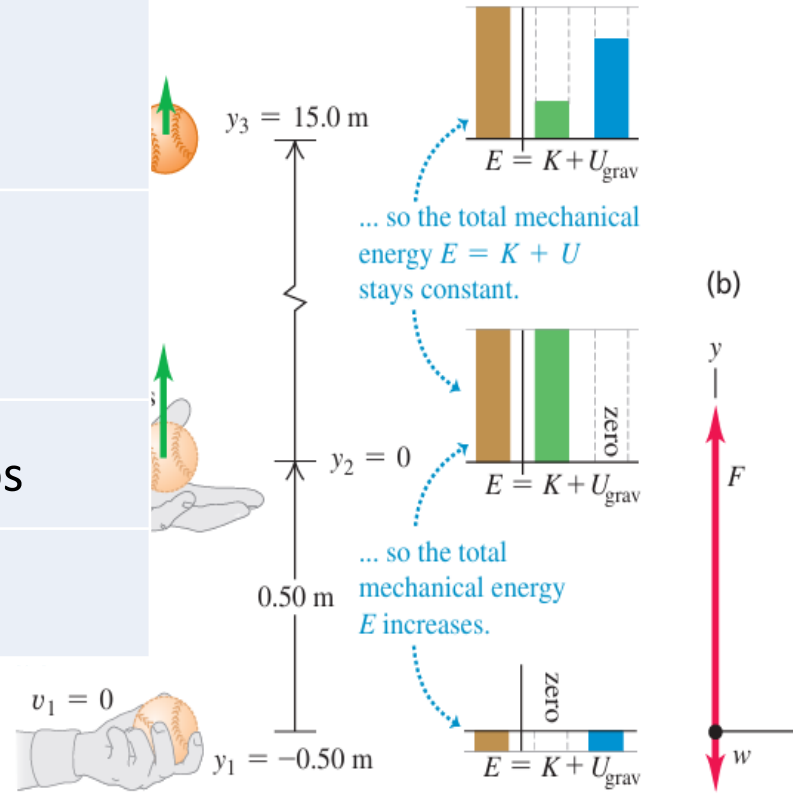
This is the single most important equation. Who write $F_{\text{hand}} = ma$(forgetting mg) get the wrong answer.

Energy item	Value	Meaning
Kinetic energy gained	$\frac{1}{2}(0.145)(20)^2 = 29.0\ \text{J}$	Speed
Potential energy gained (during throw)	$(0.145)(9.8)(0.50) = 0.71\ \text{J}$	Height
Total work done by hand	$29.0 + 0.71 = 29.71\ \text{J}$	Both jobs
Force from hand	$29.71/0.50 = 59.4\text{N}$	—



You throw a **0.145-kg** baseball straight up, suppose your hand moves **upward by 0.50 m** while you are throwing the ball. The ball leaves your hand with an **upward velocity of 20.0 m/s**. **(a) Find the magnitude of the force** (assumed constant) that your hand exerts on the ball. **(b) Find the speed of the ball at a point 15.0 m** above the point where it leaves your hand. Ignore air resistance.

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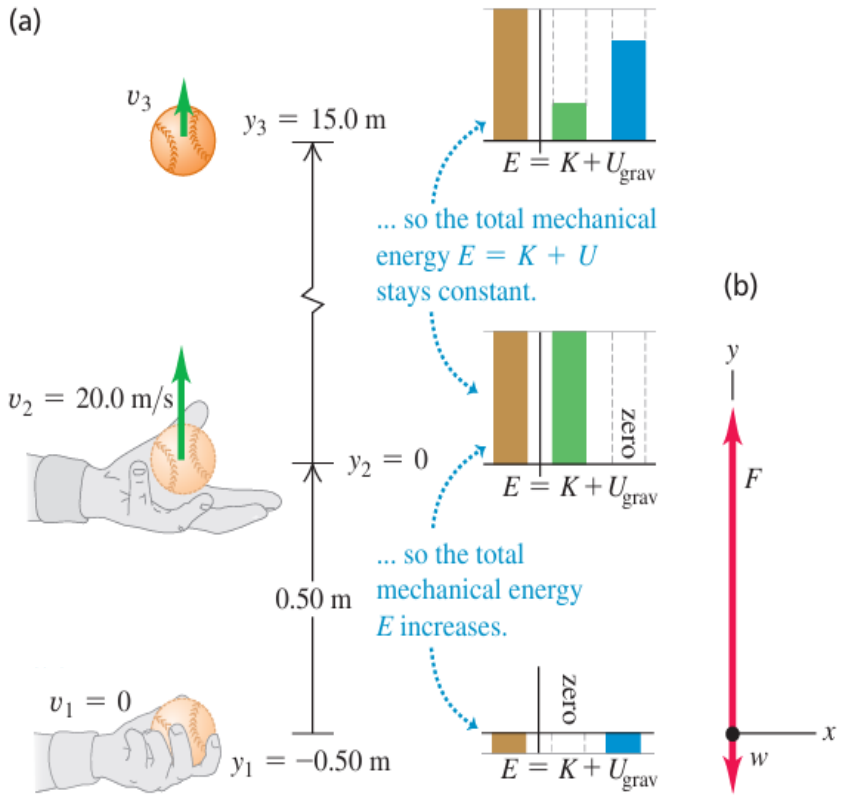
The 0.71 J is the "loss" — the energy spent lifting the ball against gravity while throwing it.

You throw a **0.145-kg** baseball straight up, suppose your hand moves **upward by 0.50 m** while you are throwing the ball. The ball leaves your hand with an **upward velocity of 20.0 m/s**. **(a) Find the magnitude of the force** (assumed constant) that your hand exerts on the ball. **(b) Find the speed of the ball at a point 15.0 m** above the point where it leaves your hand. Ignore air resistance.

$$\underbrace{F \cdot d}_{\text{work by hand}} = \underbrace{\frac{1}{2}mv^2}_{\text{KE gained}} + \underbrace{mgh}_{\text{PE gained}}$$

$$F(0.50) = 29.0 + 0.71$$

$$F = 59.4 \text{ N}$$



You throw a **0.145-kg** baseball straight up, suppose your hand moves **upward by 0.50 m** while you are throwing the ball. The ball leaves your hand with an **upward velocity of 20.0 m/s**. **(a) Find the magnitude of the force** (assumed constant) that your hand exerts on the ball. **(b) Find the speed of the ball at a point 15.0 m** above the point where it leaves your hand. Ignore air resistance.

$$K_1 = 0$$

$$U_1 = mgy_1 = (0.145)(9.8)(-0.50) = -0.710\ J$$

$$K_2 = \frac{1}{2}mv_2^2 = (1/2)(0.145)(20)^2 = 29.0\ J$$

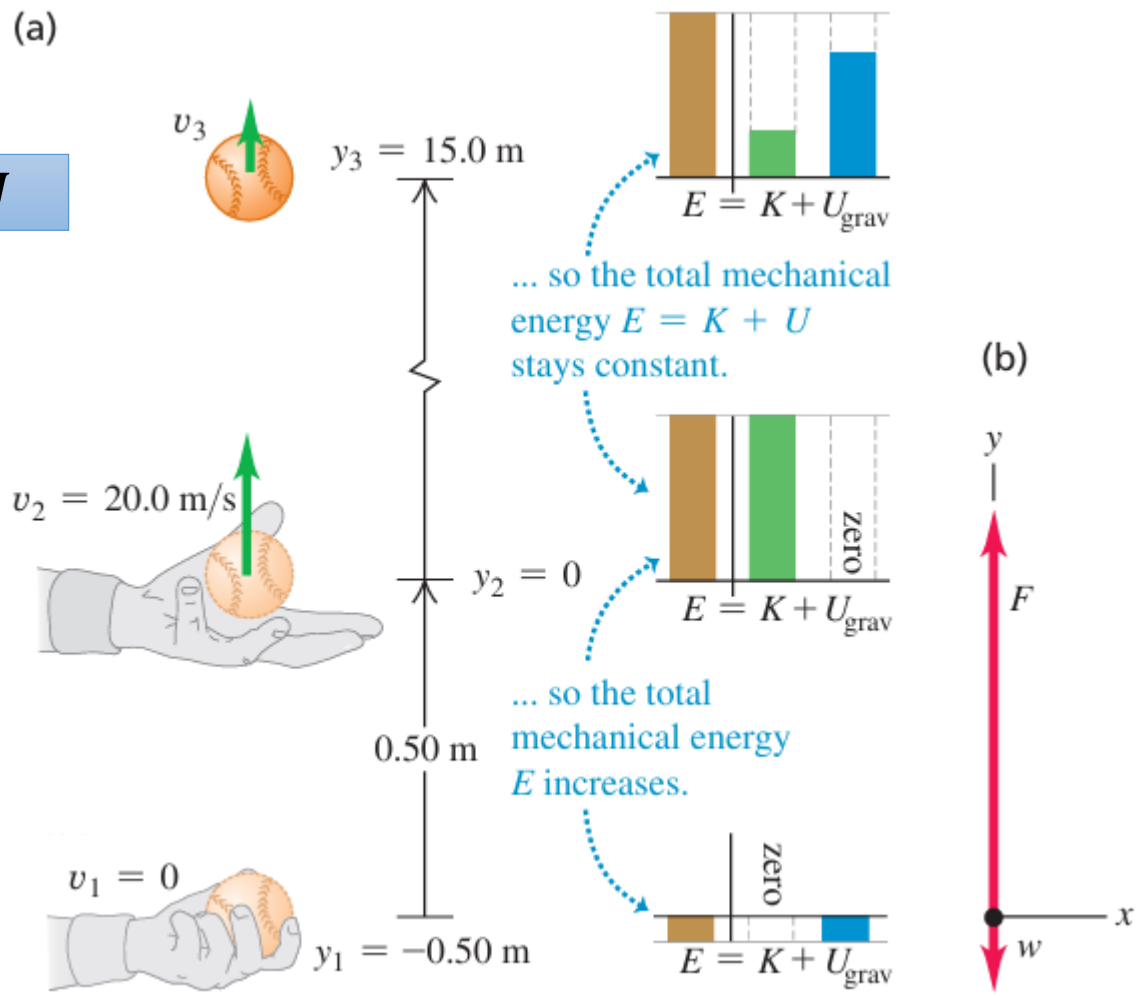
$$U_2 = mgy_2 = (0.145)(9.8)(0) = 0\ J$$

$$W_{other} + K_1 + U_1 = K_2 + U_2$$

$$W_{other} = (K_2 - K_1) + (U_2 - U_1)$$

$$W_{other} = 29.7\ J$$

$$\mathrm{F} = \frac{W_{other}}{y_2 - y_1} = \frac{29.7}{0.50} = 59.4\ N$$



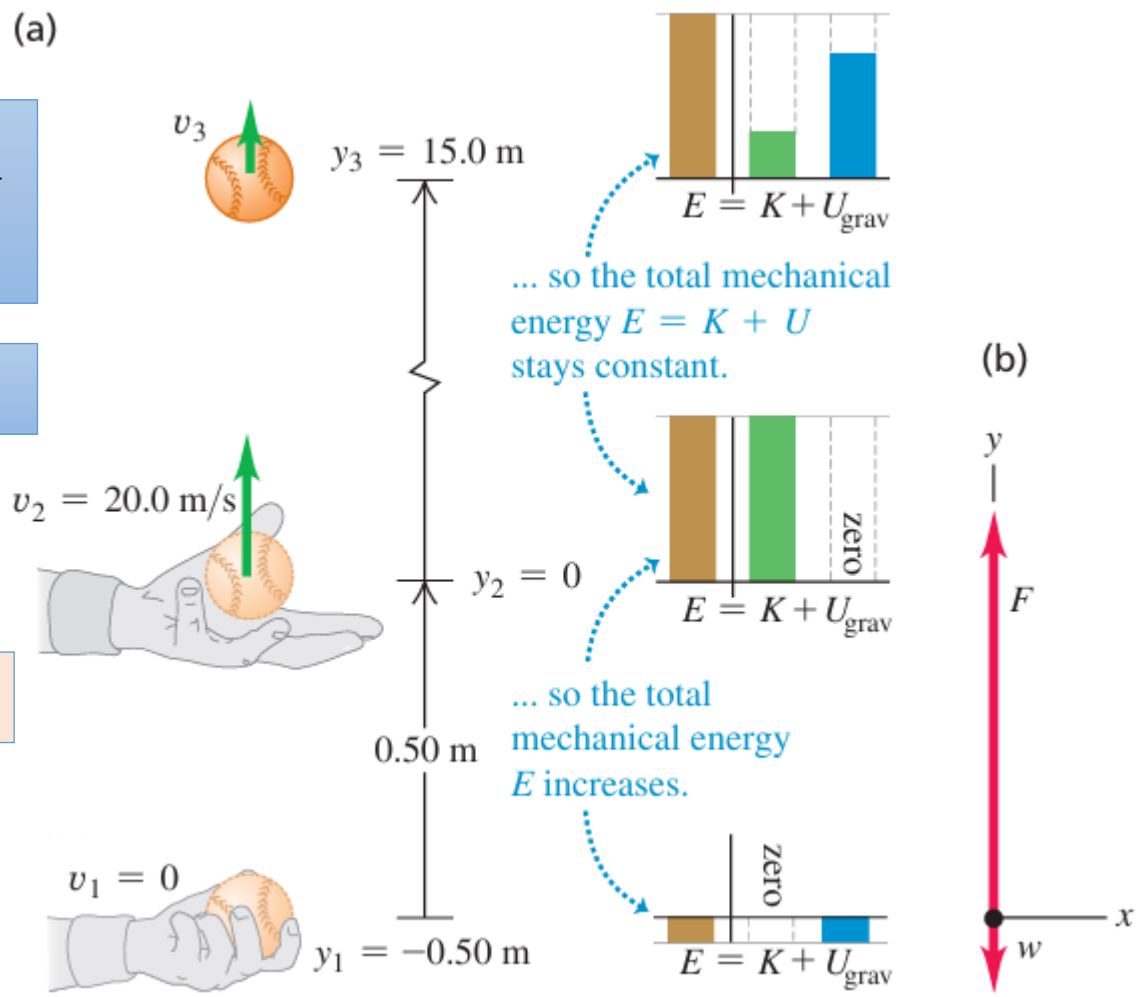
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$$K_2 = \frac{1}{2}mv_2^2 = \left(\frac{1}{2}\right)(0.145)(20)^2 = 29.0\ J$$

$$U_2 = mgy_2 = (0.145)(9.8)(0) = 0\ J$$

$$U_3 = mgy_3 = (0.145)(9.8)(15) = 21.3\ J$$

$$K_3 = ?$$



You throw a **0.145-kg** baseball straight up, suppose your hand moves **upward by 0.50 m** while you are throwing the ball. The ball leaves your hand with an **upward velocity of 20.0 m/s**. **(a) Find the magnitude of the force** (assumed constant) that your hand exerts on the ball. **(b) Find the speed of the ball at a point 15.0 m** above the point where it leaves your hand. Ignore air resistance.

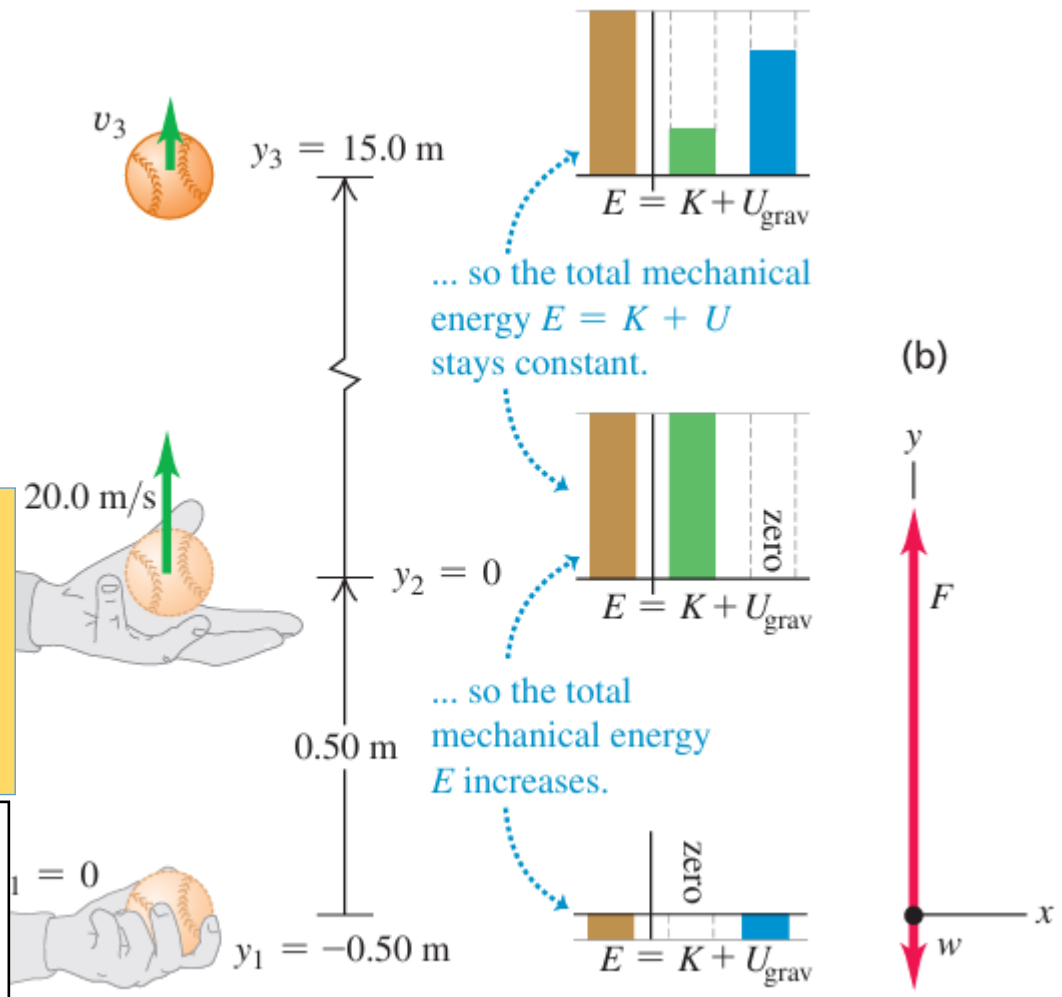
$$K_2 + U_2 = K_3 + U_3$$

$$K_3 = (K_2 + U_2) - U_3$$

$$K_3 = (29 + 0) - 21.3 = 7.7\ \text{J}$$

$$v_3 = \pm\sqrt{\frac{2K_3}{m}} = \pm\sqrt{\frac{2(7.7)}{0.145}} = \pm 10\ m/s$$

The velocity v_3 is positive +10 m/s when the ball is moving up and negative -10 m/s when it is moving down; the speed v_3 is 10 m/s in either case.

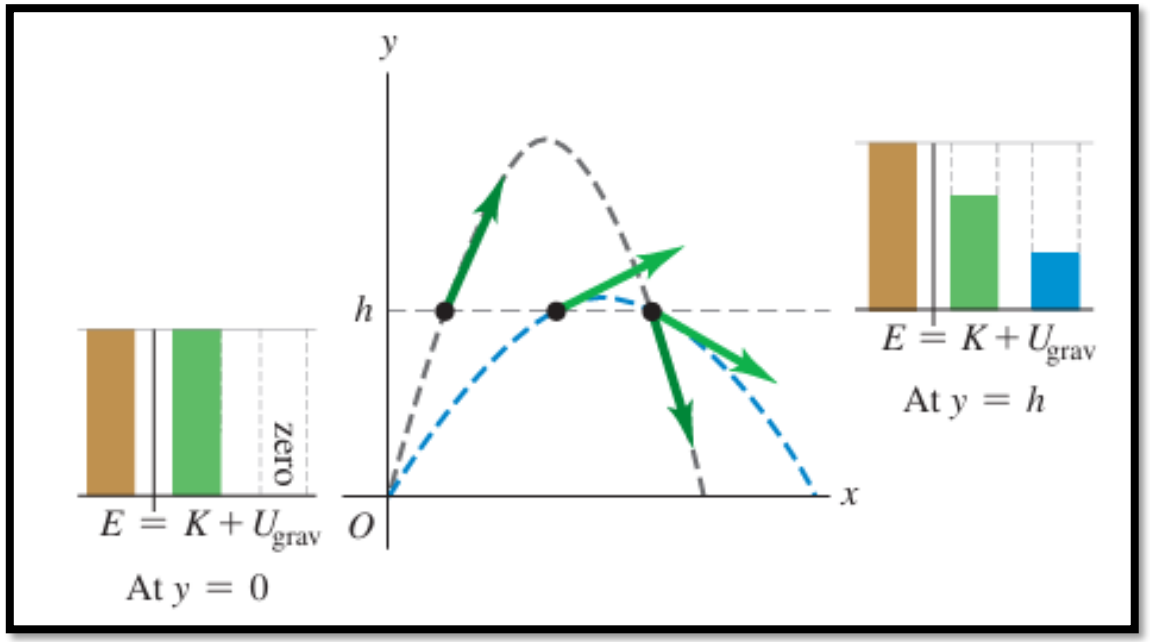


Energy in Projectile Motion

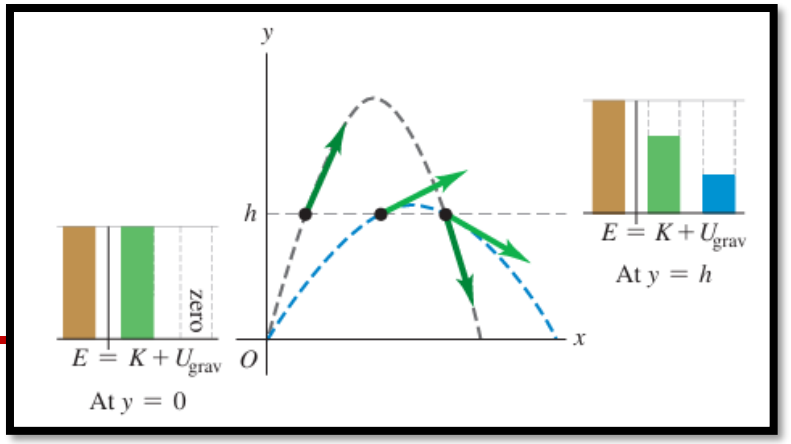
A batter hits two identical baseballs with the same initial speed and from the same initial height but at different initial angles. Prove that both balls have the same speed at any height "h" if air resistance can be ignored.

Two identical baseballs are hit with:

- **Same initial speed** v_0
- **Same initial height** (take it as $y = 0$)
- **Different launch angles** $\alpha_1 \neq \alpha_2$



Energy in Projectile Motion



$$E = K + U = \frac{1}{2}mv^2 + mgy$$

$$E_{\text{initial}} = E_{\text{final}}$$

$$\frac{1}{2}mv_0^2 + mgy_0 = \frac{1}{2}mv^2 + mgh$$

$$\frac{1}{2}v^2 = \frac{1}{2}v_0^2 + gy_0 - gh$$

$$v^2 = v_0^2 + 2g(y_0 - h)$$

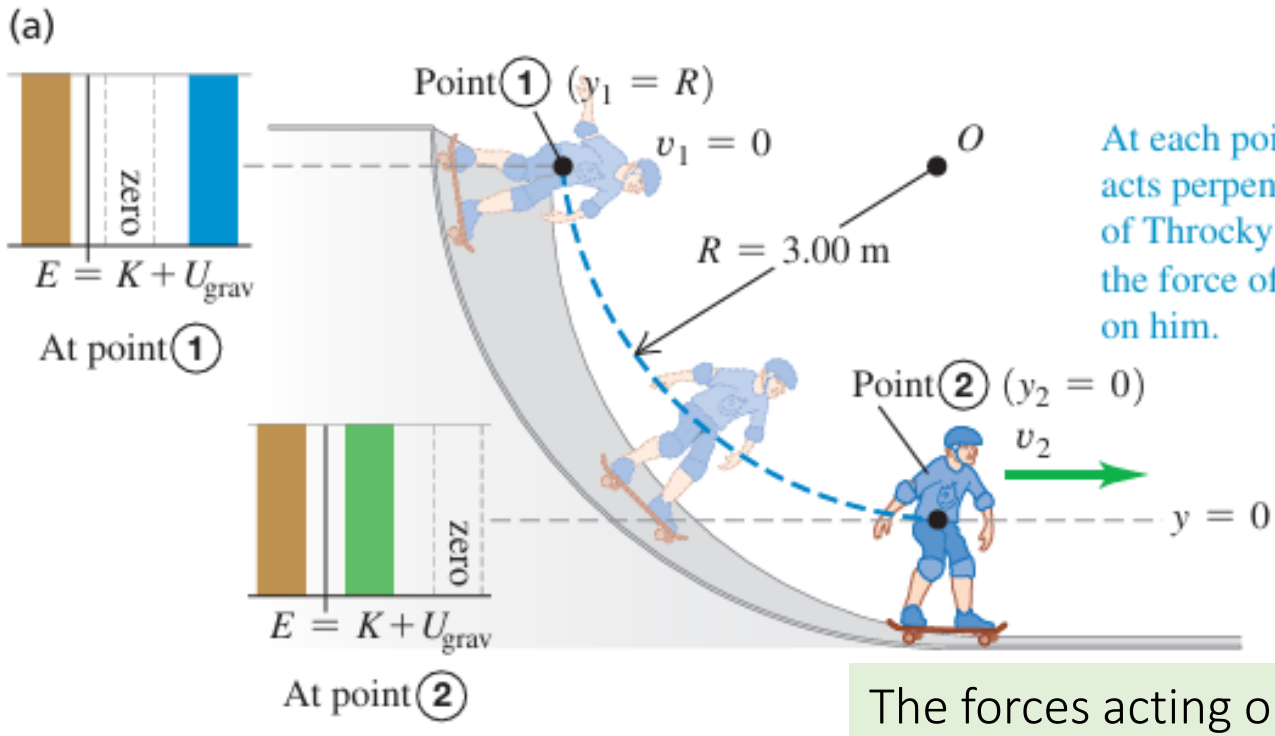
$$v = \sqrt{v_0^2 + 2g(y_0 - h)}$$

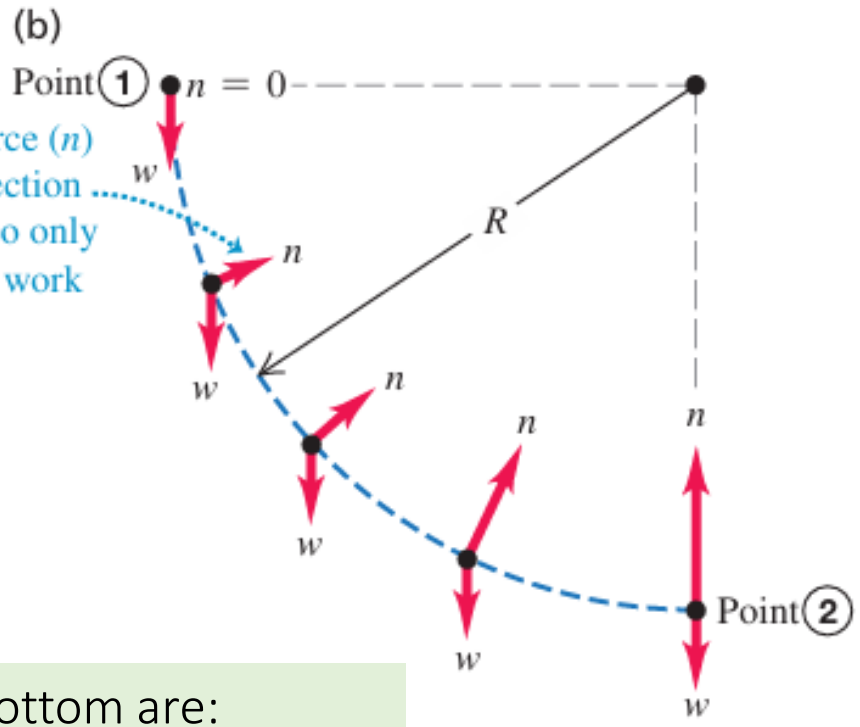
The launch angle (θ) **does not appear anywhere in the final equation**.

The speed v at any height h depends solely on:

1. The initial speed (v_0)
2. The initial height (y_0)
3. Acceleration due to gravity (g)
4. The target height (h)

Your cousin Ali skateboards from rest down a curved, frictionless ramp. If we treat Ali and his skateboard as a particle, he moves through a quarter-circle with **radius $R = 3.00$ m.** Ali and his skateboard have a **total mass of 25.0 kg**. **(a) Find his speed at the bottom** of the ramp. **(b) Find the normal force** that acts on him at the bottom of the curve.

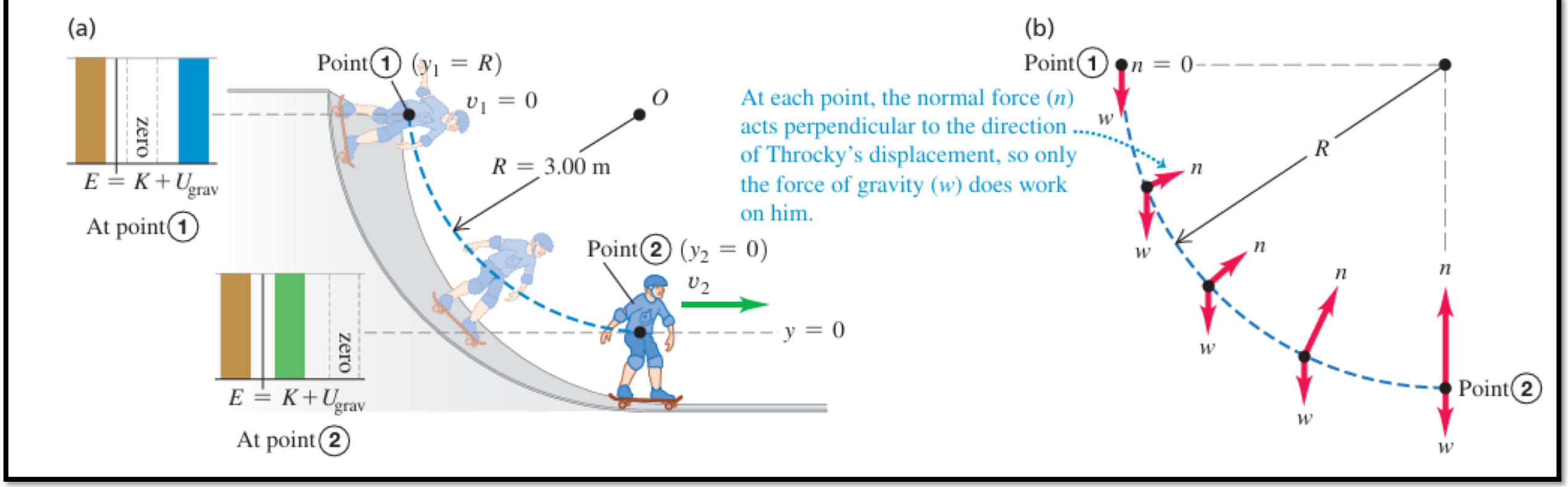




At each point, the normal force (n) acts perpendicular to the direction of Throcky's displacement, so only the force of gravity (w) does work on him.

The forces acting on Ali at the bottom are:

1. **Gravity** mg — downward
2. **Normal force** N — upward (from the ramp)



$$KE_1 = 0$$

$$KE_2 = \frac{1}{2}mV_2^2$$

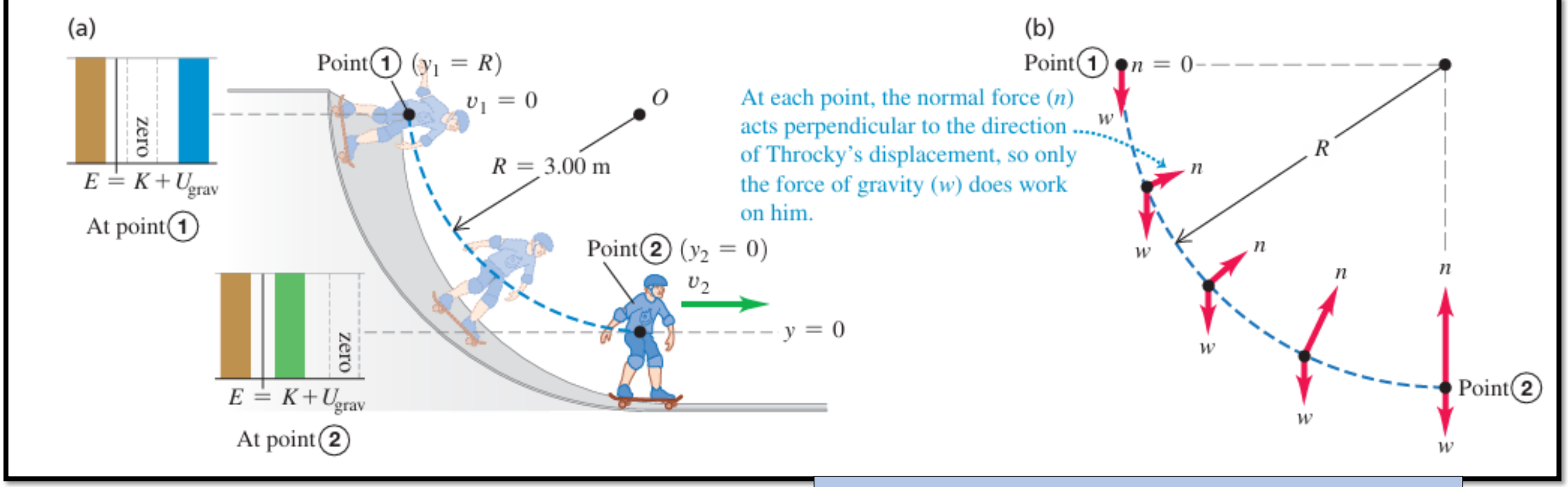
$$U_1 = mgR$$

$$U_2 = 0$$

$$K_1 + U_1 = K_2 + U_2$$

$$v_2 = \sqrt{2gR}$$

$$v_2 = 7.67\ m/s$$



$$a_{\text{rad}} = \frac{V_2^2}{R}$$

$$n = mg + m\frac{v^2}{R}$$

$$\sum F_y = n + (-w)$$

As $v_2 = \sqrt{2gR}$

$$ma_{rad} = n + (-w)$$

$$a_{\text{rad}} = \frac{2gR}{R} = 2g$$

$$m(2g) = n + (-w)$$

$$n = 2mg + (w) = 3mg = 735\ N$$

A **2000-kg (119,600 N)** elevator with
contacts a cushioning spring at the b
elevator, **compressing 2.00 m** as it d
constant 17,000-N friction force to
the spring?

$$m = 2000\ kg$$

$$V_1 = 4\ m/s$$

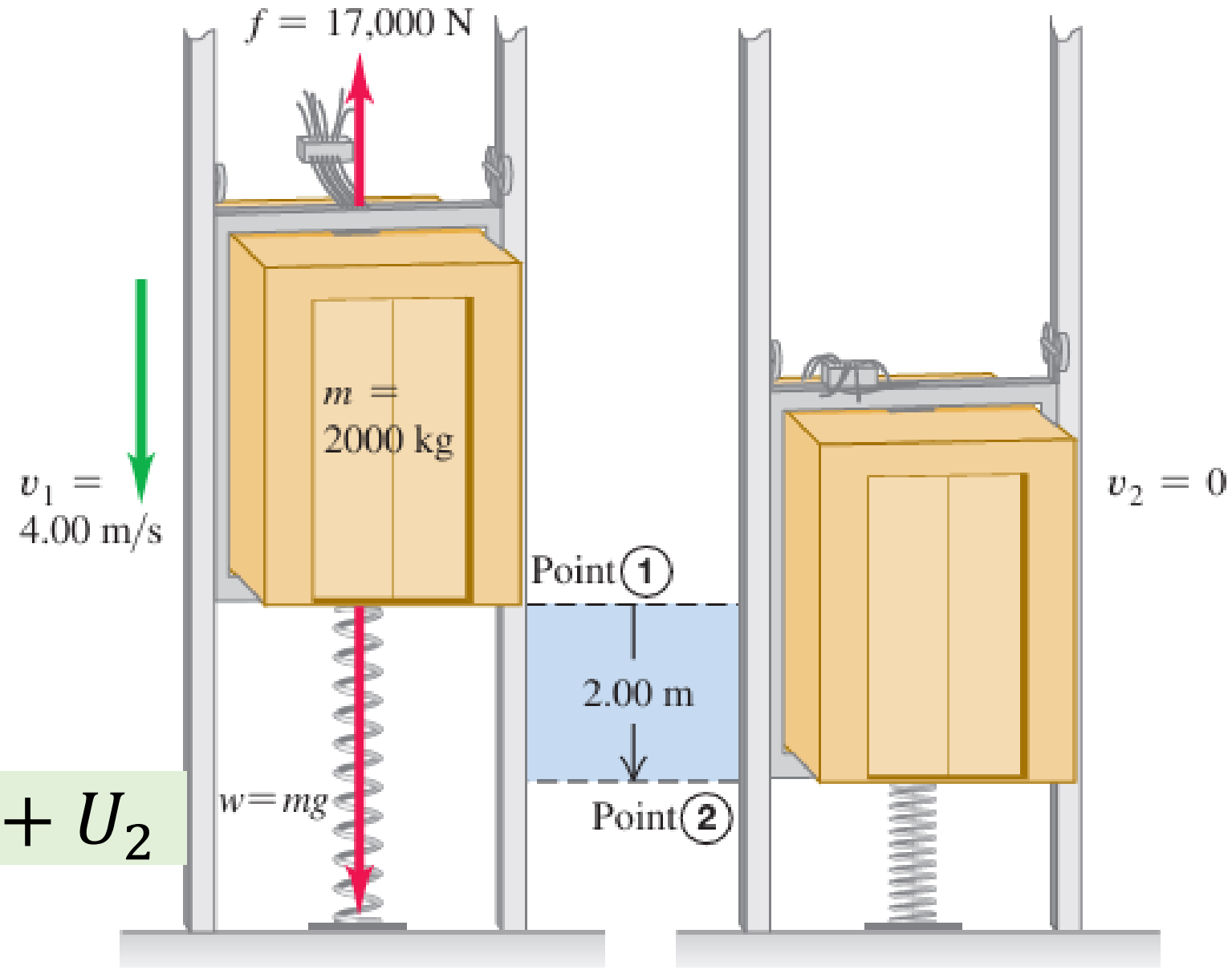
$$f = 17000\ \text{N}$$

$$K = ?$$

$$K_1 + U_1 = K_2 + U_2$$

$$K_1 + U_1 + W_{other} = K_2 + U_2$$

Figure 7.17 The fall of an elevator is stopped by a spring and by a constant friction force.



A **2000-kg (119,600 N)** elevator with broken cables in a test rig is falling at **4.00 m/s** when it contacts a cushioning spring at the bottom of the shaft. The spring is intended to stop the elevator, **compressing 2.00 m** as it does so (**Fig.**). During the motion a safety clamp applies a **constant 17,000-N friction force** to the elevator. **What is the necessary force constant *k* for the spring?**

$$K_1 = \frac{1}{2}mv_{1y}^2 = \frac{1}{2}(2000)(4)_{1y}^2 = 16000\,J$$

$$U_1 = U_{1,grav} + U_{1,elastic} = mh(0) + \frac{1}{2}K(0)_1^2 = 0\,J$$

$$K_2 = \frac{1}{2}mv_{2y}^2$$

$$U_2 = U_{2,grav} + U_{2,elastic} = mhy_2 + \frac{1}{2}Ky_2^2 = (2000)(9.8)(-2) + \frac{1}{2}Ky_2^2$$

$$U_2 = -(39200) + \frac{1}{2}Ky_2^2$$

A **2000-kg (119,600 N)** elevator with broken cables in a test rig is falling at **4.00 m/s** when it contacts a cushioning spring at the bottom of the shaft. The spring is intended to stop the elevator, **compressing 2.00 m** as it does so (**Fig.**). During the motion a safety clamp applies a **constant 17,000-N friction force** to the elevator. **What is the necessary force constant *k* for the spring?**

$$K_1 + U_1 + W_{other} = K_2 + U_2$$

$$W_{other} = -(17000)(2) = -34000$$

$$16000 + 0 + (-34000) = -(39200) + \frac{1}{2}Ky_2^2$$

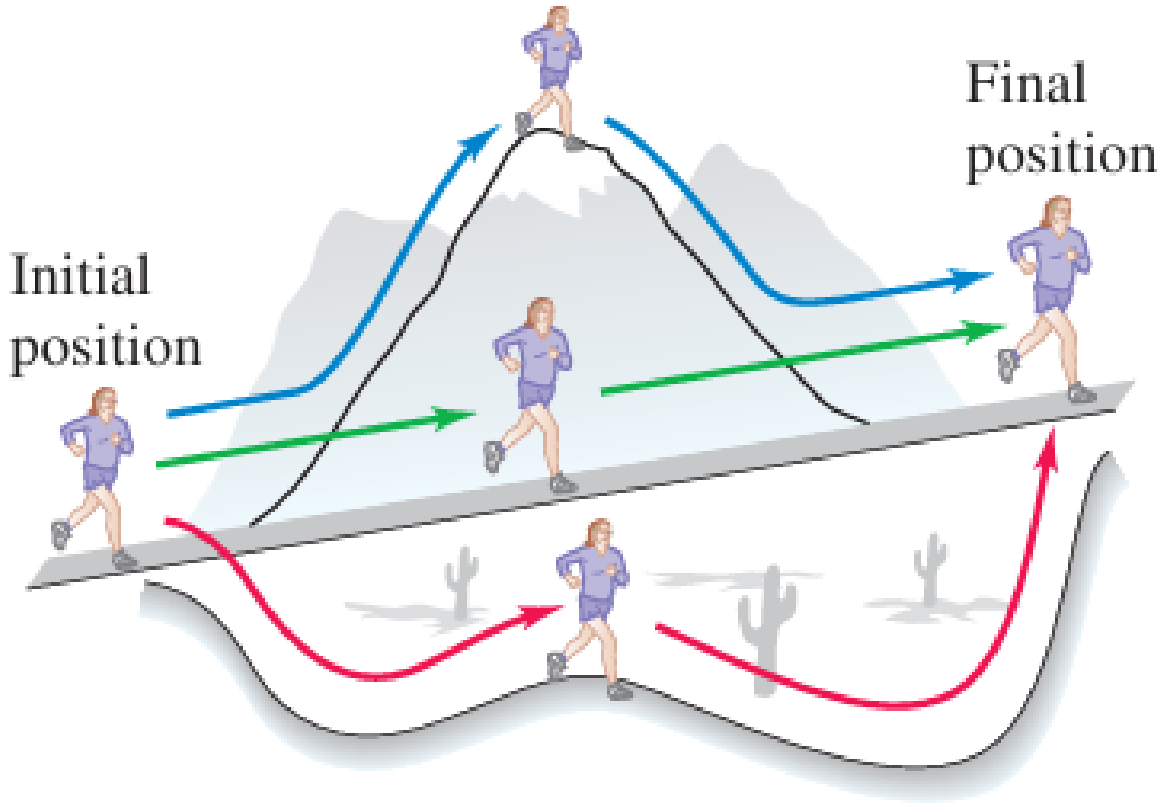
$$16000 + 0 + (-34000) = -(39200) + \frac{1}{2}K(-2)_2^2$$

$$K = 1.06 \times 10^4 N/m$$

Conservative and Non-conservative Forces

The work done by a conservative force such as gravity depends on only the endpoints of a path, not the specific path taken between those points.

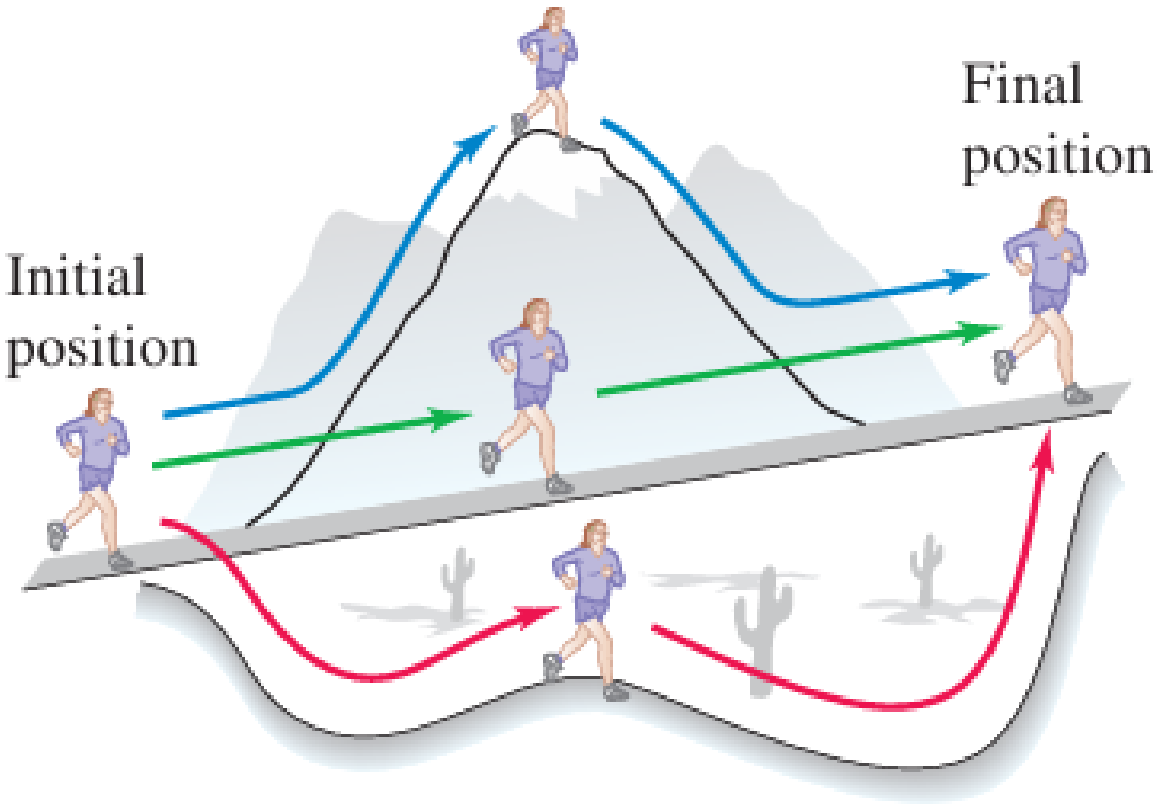
The gravitational force is conservative, the work it does is the same for all three paths



Conservative and Non-conservative Forces

Work done is path independent

Work done along close path is zero



Conservative and Non-conservative Forces

1.Path Dependence:

•Conservative Forces: Work done is the same regardless of the path taken between two points.

•Non-Conservative Forces: Work done depends on the specific path taken.

2.Energy Conservation:

•Conservative Forces: Total mechanical energy (kinetic + potential) remains constant in a closed s ystem.

•Non-

Conservative Forces: Mechanical energy is not conserved; some energy is lost as heat or sound.

3.Potential Energy:

•Conservative Forces: Can be associated with a potential energy function (e.g., gravitational pote ntial energy).

•Non-Conservative Forces: Cannot be associated with potential energy since energy is dissipated.

Conservative and Non-conservative Forces

Conservative Forces:

Gravitational Force: The work done in lifting an object is independent of the path ta ken.

Spring Force: The work done in compressing or stretching a spring is path-independent.

Non-Conservative Forces:

Friction: The work done against friction depends on the distance traveled and the p ath taken.

Air Resistance: The energy lost due to air resistance varies with speed and distance traveled.

Conservative and Non-conservative Forces

APPLICATION Nonconservative Forces and Internal Energy in a Tire

An automobile tire deforms and flexes like a spring as it rolls, but it is not an ideal spring: Nonconservative internal friction forces act within the rubber. As a result, mechanical energy is lost and converted to internal energy of the tire. Thus the temperature of a tire increases as it rolls, which causes the pressure of the air inside the tire to increase as well. That's why tire pressures are best checked before the car is driven, when the tire is cold.



Potential Energy and Kinetic Energy

$$K_1 + U_1 = K_2 + U_2$$

$$K_1 + U_{g,1} = K_2 + U_{g,2}$$

$$K_1 + U_{e,1} = K_2 + U_{e,2}$$

$$K_1 + U_{g,1} + U_{e,1} = K_2 + U_{g,2} + U_{e,2}$$

$$W_{other} + K_1 + U_{g,1} + U_{e,1} = K_2 + U_{g,2} + U_{e,2}$$

$$K_1 = \frac{1}{2}mv_{1x}^2 \qquad K_2 = \frac{1}{2}mv_{2x}^2$$

$$U_{e,1} = \frac{1}{2}Kx_1^2 \qquad U_{e,2} = \frac{1}{2}Kx_2^2$$

$$U_{g,1} = mgh_1 \qquad U_{g,2} = mgh_2$$

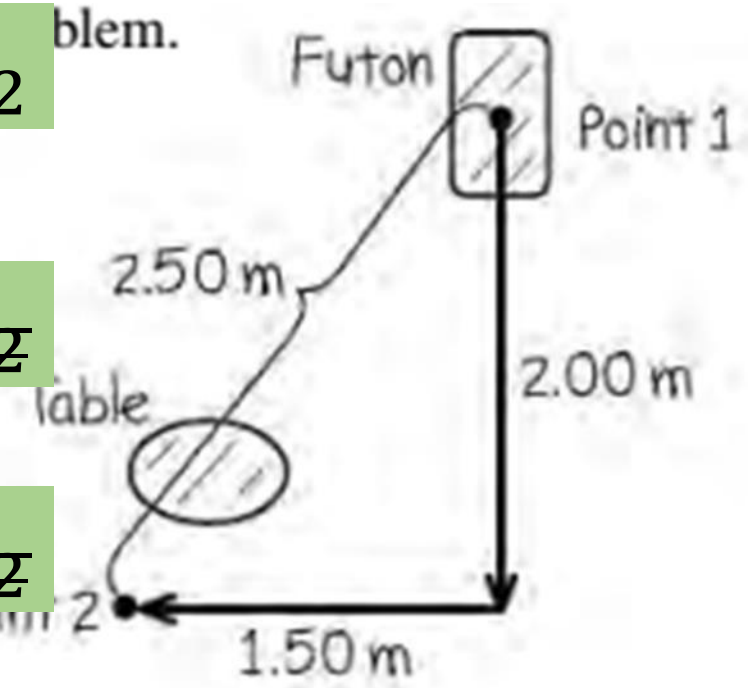
You are rearranging your furniture and wish to move a **40.0-kg futon** **2.50 m across the room**. A heavy coffee table, which you don't want to move, blocks this straight-line path. Instead, you slide the futon along a dogleg path; **the doglegs are 2.00 m and 1.50 m long**. **How much more work must you** do to push the futon along the dogleg path than along the straight-line path? The coefficient of **kinetic friction** is $\mu_k = 0.200$.

$$W_{other} + K_1 + U_{g,1} + U_{e,1} = K_2 + U_{g,2} + U_{e,2}$$

$$W_{other} + K_1 + U_{g,1} + \cancel{U_{e,1}} = K_2 + U_{g,2} + \cancel{U_{e,2}}$$

$$W_{other} + K_1 + \cancel{U_{g,1}} + \cancel{U_{e,1}} = K_2 + \cancel{U_{g,2}} + \cancel{U_{e,2}}$$

$$W_{other} + \cancel{K_1} + \cancel{U_{g,1}} + \cancel{U_{e,1}} = \cancel{K_2} + \cancel{U_{g,2}} + \cancel{U_{e,2}}$$







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$$W_{other} + \cancel{K_1} + \cancel{U_{g,1}} + \cancel{U_{e,1}} = \cancel{K_2} + \cancel{U_{g,2}} + \cancel{U_{e,2}}$$

$$W_{you} + W_f = 0$$

$$W_{you} = -W_f$$

$$W_{you} = -(-f_K d) = f_K d = (\mu_K n)d$$

$$W_{you} = \mu_K (mg)d$$

Force and Potential Energy

Gravitational Force

$$F_g = -mg$$

$$U_g = mgy$$

$$W = -\Delta U$$

$$F_x \Delta x = -\Delta U$$

$$F_x = -\frac{\Delta U}{\Delta x}$$

$$F_x = -\lim_{\Delta x \to 0} \frac{\Delta U}{\Delta x} = -\frac{dU}{dx}$$

Force and Potential Energy

$$F_x = -\frac{dU}{dx}$$

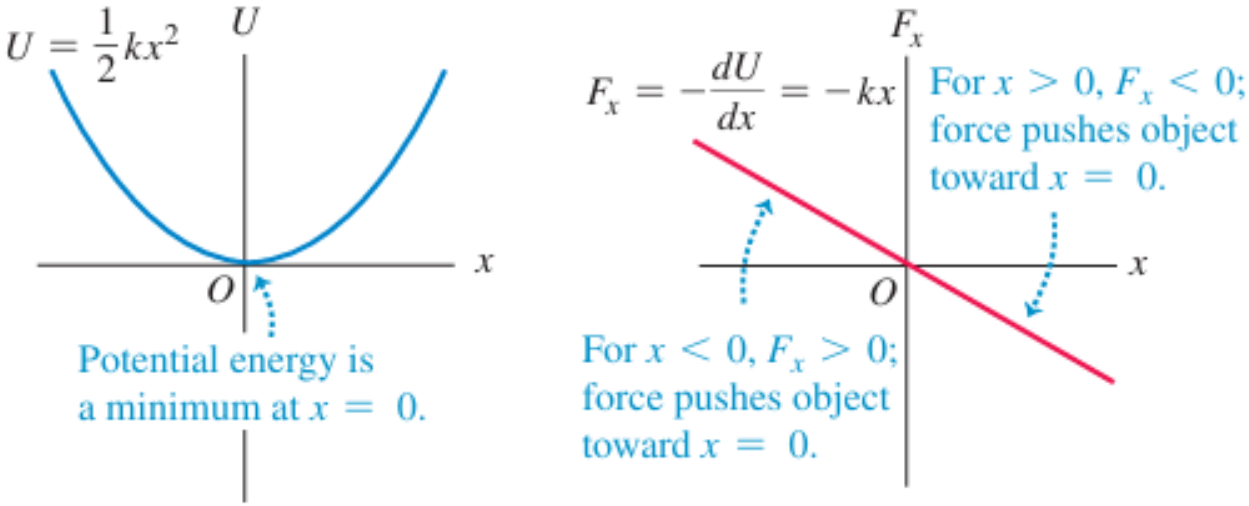
$$U = \frac{1}{2}Kx_1^2$$

$$F_x = -\frac{d\left(\frac{1}{2}Kx_1^2\right)}{dx}$$

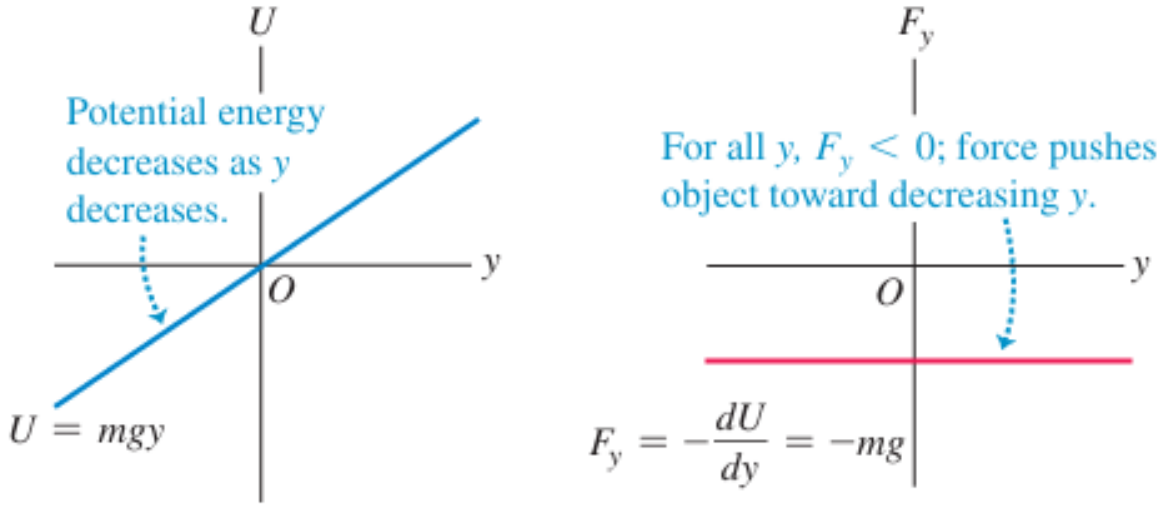
$$F_x = -Kx$$

Figure 7.22 A conservative force is the negative derivative of the corresponding potential energy.

(a) Elastic potential energy and force as functions of x



(b) Gravitational potential energy and force as functions of y



An electrically charged particle is **held at rest at the point $x = 0$**; a second particle with equal charge is free to move along the positive x-axis. **The potential energy of the system is $U(x) = C/x$**, where C is a positive constant that depends on the magnitude of the charges. **Derive an expression for the x-component of force** acting on the movable particle as a function of its position.

$$F_x = -\frac{d(C/x)}{dx}$$

$$F_x = -C\frac{d(1/x)}{dx}$$

$$F_x = -C(-x^{-2})$$

$$F_x = +C(x^{-2})$$

$$F_x = -\frac{dU}{dx}$$

Force and PE: Three Dimensions

$$F_x = -\frac{dU}{dx}$$

$$F_y = -\frac{dU}{dy}$$

$$F_z = -\frac{dU}{dz}$$

$$\vec{F} = -\left(F_x\hat{\imath} + F_y\hat{\jmath} + F_z\hat{k}\right)$$

$$\vec{F} = -\left(\frac{\partial U}{\partial x}\hat{\imath} + \frac{\partial U}{\partial y}\hat{\jmath} + \frac{\partial U}{\partial z}\hat{k}\right)$$

$$\vec{F} = -\left(\frac{\partial}{\partial x}\hat{\imath} + \frac{\partial}{\partial y}\hat{\jmath} + \frac{\partial}{\partial z}\hat{k}\right)U$$

$$\vec{F} = -\bar{\nabla}U$$

A puck with coordinates x and y slides on a level, frictionless air-hockey table. It is acted on by a conservative force described by the potential-energy function

$$U(x, y) = \tfrac{1}{2}k(x^2 + y^2)$$

Note that $r = \sqrt{x^2 + y^2}$ is the distance on the table surface from the puck to the origin. Find a vector expression for the force acting on the puck, and find an expression for the magnitude of the force.

$$\vec{F} = -\left(\frac{\partial}{\partial x}\hat{\imath} + \frac{\partial}{\partial y}\hat{\jmath} + \frac{\partial}{\partial z}\hat{k}\right)U$$

$$\vec{F} = -\left(\frac{\partial}{\partial x}\hat{\imath} + \frac{\partial}{\partial y}\hat{\jmath} + \frac{\partial}{\partial z}\hat{k}\right)\left(\frac{1}{2}k\right)(y^2 + x^2)$$

Example

$$\vec{F} = -\left(\frac{\partial}{\partial x}\hat{\imath} + \frac{\partial}{\partial y}\hat{\jmath} + \frac{\partial}{\partial z}\hat{k}\right)U$$

$$\vec{F} = -\left(\frac{\partial}{\partial x}\left[\left(\frac{1}{2}k\right)(y^2 + x^2)\right]\hat{\imath} + \frac{\partial}{\partial y}\left[\left(\frac{1}{2}k\right)(y^2 + x^2)\right]\hat{\jmath} + \frac{\partial}{\partial z}\hat{k}\right)$$

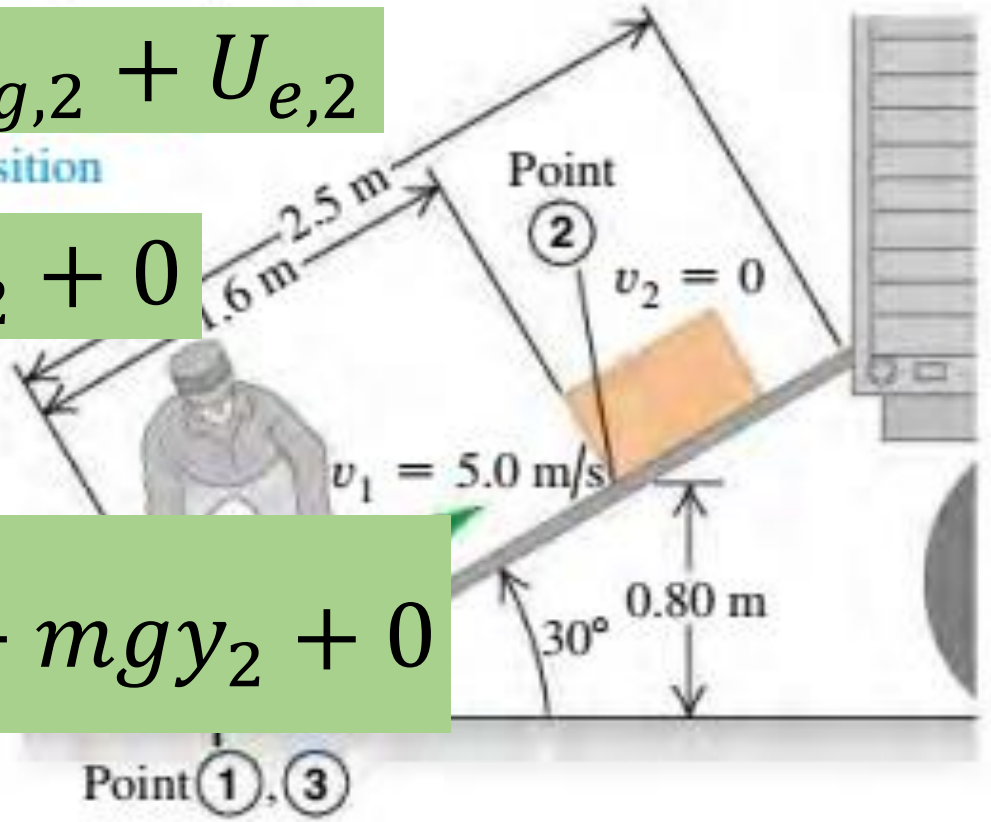
$$\vec{F} = -kx\hat{\imath} - ky\hat{\jmath} + 0\hat{k}$$

We want to slide a **12-kg crate** up a **2.5-m-long ramp inclined at 30°.** A worker, ignoring friction, calculates that he can do this by giving it an **initial speed of 5.0 m/s** at the bottom and letting it go. But friction is *not* negligible; **the crate slides only 1.6 m up** the ramp, stops, and slides back down (Fig.) **(a) Find the magnitude of the friction force** acting on the crate, assuming that it is constant.

$$W_{other} + K_1 + U_{g,1} + U_{e,1} = K_2 + U_{g,2} + U_{e,2}$$

$$W_{other} + K_1 + U_{g,1} + 0 = K_2 + U_{g,2} + 0$$

$$fd + \frac{1}{2}mv_1^2 + mgy_1 + 0 = \frac{1}{2}mv_2^2 + mgy_2 + 0$$

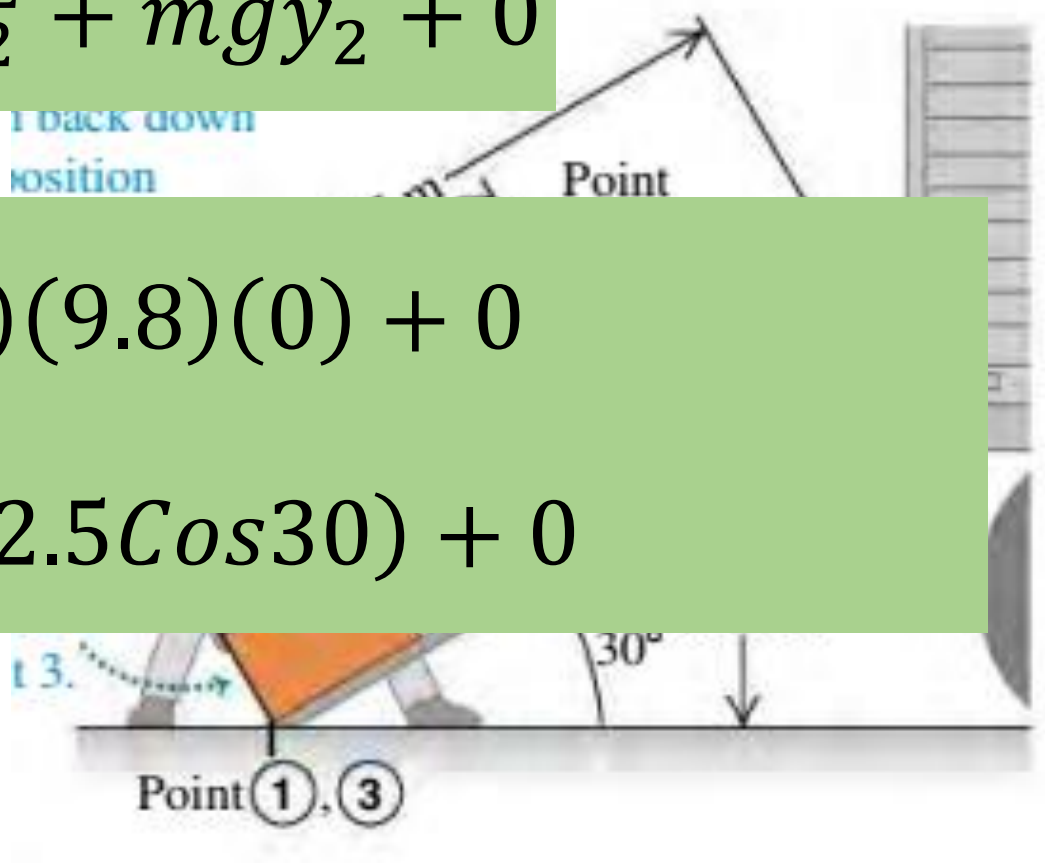


Example

$$fd + \frac{1}{2}mv_1^2 + mgy_1 + 0 = \frac{1}{2}mv_2^2 + mgy_2 + 0$$

$$f(1.6) + \frac{1}{2}m(5)_1^2 + (12)(9.8)(0) + 0$$

$$= \frac{1}{2}m(0)_2^2 + (12)(9.8)(2.5Cos30) + 0$$



Energy Diagram

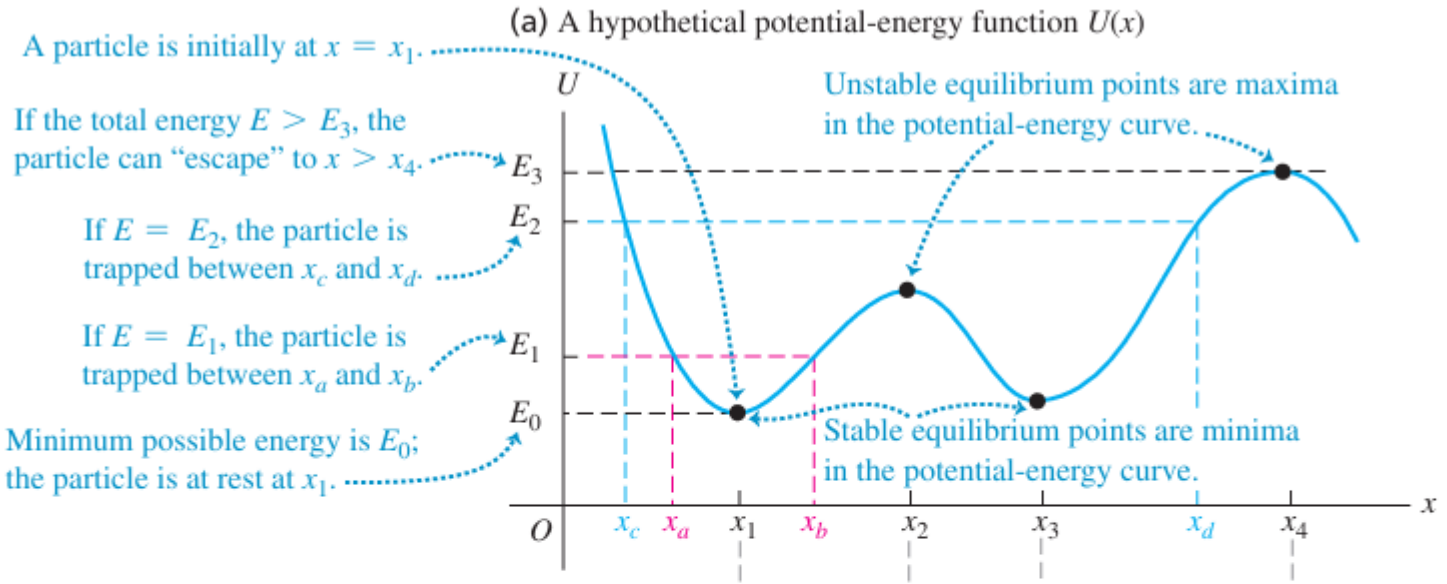
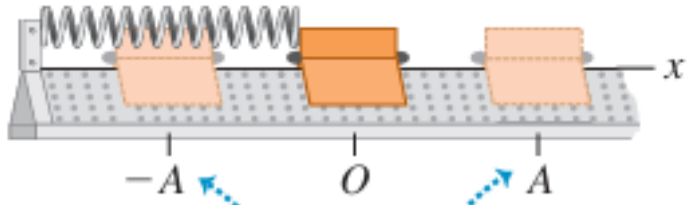


Figure 7.23 (a) A glider on an air track. The spring exerts a force $F_x = -kx$. (b) The potential-energy function.

(a)



The limits of the glider's motion are at $x = A$ and $x = -A$.

(b)

On the graph, the limits of motion are the points where the U curve intersects the horizontal line representing total mechanical energy E.

